

BAYRAMALY, TurkmenistanWMO: **388950**Lat: **37.600N**Lon: **62.183E**Elev: **240**StdP: **98.47**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-5.0	-14.3	1.1	-5.0	-11.0	1.5	-0.4	7.3	14.3	6.3	12.1	1.8	80	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	40.9	21.8	39.5	21.3	38.2	20.8	23.7	37.6	22.7	36.8	21.8	36.1	2.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.6	13.8	33.7	17.2	12.7	31.8	16.2	11.8	30.7	72.1	37.4	68.1	36.9	64.9	36.2	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.8	4.9	4.2	-10.6	43.3	4.2	1.2	-13.5	44.1	-16.0	44.8	-18.3	45.4	-21.4	46.2		
(5)			-11.1	25.2	4.1	1.2	-14.1	26.1	-16.5	26.8	-18.8	27.5	-21.8	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	4.9	6.6	12.6	18.8	25.0	29.6	31.5	29.6	23.8	17.2	10.3	5.8	(6)
(7)		DBStd	10.37	5.29	5.49	5.04	4.90	3.48	2.68	2.35	2.71	3.37	4.51	5.04	4.65	(7)
(8)		HDD10.0	527	171	117	32	3	0	0	0	0	0	4	55	144	(8)
(9)		HDD18.3	1698	418	328	187	54	2	0	0	0	3	75	243	388	(9)
(10)		CDD10.0	3460	12	23	111	266	464	588	666	609	415	227	64	15	(10)
(11)		CDD18.3	1591	0	0	9	67	209	338	407	351	168	39	3	0	(11)
(12)		CDH23.3	21551	1	5	116	699	2476	4867	6161	4812	1910	459	44	1	(12)
(13)		CDH26.7	12516	0	1	29	282	1257	2969	3947	2941	928	151	11	0	(13)
(14)	Wind	WSAvg	2.1	2.3	2.4	2.4	2.2	2.0	2.0	1.9	1.8	1.8	1.9	2.2	(14)	
(15)	Precipitation	PrecAvg	178	30	25	36	29	15	1	0	1	4	15	24	(15)	
(16)		PrecMax	274	109	108	97	82	63	10	0	2	14	28	79	62	(16)
(17)		PrecMin	72	1	2	8	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	46	26	21	19	22	16	2	0	0	2	7	16	18	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.6	23.6	29.9	35.7	38.6	41.4	43.4	42.0	38.2	34.0	28.2	21.7	(19)
(20)			MCWB	11.2	12.5	15.6	18.8	20.6	22.0	21.8	21.8	20.0	18.5	15.4	11.1	(20)
(21)		2%	DB	17.9	20.3	26.4	32.5	36.4	39.7	41.4	40.1	35.9	30.3	23.8	18.6	(21)
(22)			MCWB	9.8	11.1	14.6	18.2	19.7	21.0	22.0	21.6	19.6	16.7	13.4	10.1	(22)
(23)		5%	DB	15.1	17.4	23.7	30.0	34.8	38.4	40.0	38.6	33.9	28.0	21.0	15.7	(23)
(24)			MCWB	8.7	10.0	13.3	16.9	19.2	20.7	21.6	20.9	18.5	16.1	12.3	9.1	(24)
(25)		10%	DB	12.2	14.8	20.9	27.5	32.9	37.1	38.5	37.0	31.9	25.8	18.6	13.0	(25)
(26)			MCWB	7.3	8.8	12.5	16.1	18.5	20.4	21.2	20.2	17.9	14.8	11.2	7.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	13.4	17.3	21.0	22.2	24.2	25.1	24.9	22.1	19.4	16.5	12.4	(27)
(28)			MCD B	18.7	20.9	27.1	32.2	35.3	37.5	39.4	39.1	35.6	31.7	26.4	19.3	(28)
(29)		2%	WB	10.3	11.9	15.4	19.2	20.9	22.7	23.9	23.2	20.3	17.7	14.4	10.7	(29)
(30)			MCD B	16.6	18.8	23.9	29.9	34.0	37.0	37.7	37.5	33.7	28.5	21.5	17.1	(30)
(31)		5%	WB	8.9	10.5	14.2	17.8	20.0	21.9	22.9	22.0	19.3	16.5	12.9	9.4	(31)
(32)			MCD B	14.1	16.3	22.2	27.4	32.2	36.2	37.2	36.1	32.3	26.6	19.8	15.0	(32)
(33)		10%	WB	7.5	9.3	13.0	16.7	19.1	21.0	22.0	21.0	18.3	15.4	11.7	8.1	(33)
(34)			MCD B	12.1	14.3	20.0	25.7	31.0	35.5	36.6	35.2	30.6	24.5	17.9	12.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	11.2	12.2	13.2	13.5	13.3	14.1	14.7	14.0	11.5	9.4	(35)
(36)			MCDBR	14.1	14.8	15.9	16.1	15.6	15.5	15.0	15.8	17.0	16.8	15.3	13.9	(36)
(37)			MCWBR	7.5	7.5	6.6	6.4	6.3	6.2	6.1	6.8	7.2	8.1	7.8	7.7	(37)
(38)		5% WB	MCD BR	12.7	13.2	14.3	14.1	14.1	14.4	14.2	14.9	15.8	15.1	13.5	12.8	(38)
(39)			MCWBR	7.2	7.1	6.4	6.2	6.1	6.6	6.8	7.4	7.4	7.6	7.3	7.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.389	0.434	0.505	0.534	0.537	0.542	0.525	0.481	0.453	0.483	0.445	0.395	(40)	
(41)		taud	2.091	1.978	1.835	1.809	1.825	1.806	1.849	1.937	1.975	1.881	1.959	2.068	(41)	
(42)		Ebn at Noon	757	769	751	758	765	760	769	794	790	704	680	713	(42)	
(43)		Edh at Noon	120	152	193	210	210	214	203	182	165	165	136	115	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.24	3.20	4.34	5.74	7.11	8.04	7.92	7.32	6.05	4.25	2.75	2.08	(44)	
(45)		RadStd	0.19	0.33	0.34	0.48	0.26	0.24	0.17	0.18	0.15	0.24	0.25	0.23	(45)	